Ironhack requirements map — Case Study 1

This project maps to the supplied **Data Science End-to-End Project** brief. It documents a predictive Python pipeline. It does not claim a guaranteed jury outcome, GitHub publication, or portal submission.

Brief requirement	Repository evidence / status	Boundary or user action
Business case and measurable target	README, methodology	Supplied-label prediction; not product-safety validation.
Data collection and understanding	`data/`, source/audit reports, `src/collection.py`	API is optional; acquisition is required. Historical labels are not regenerated.
Cleaning and wrangling	`src/cleaning.py`, cleaning report	47 rows quarantined; no missing registry is invented.
EDA and charts	`src/analysis.py`, `reports/figures/`, notebook	Descriptive results are not toxicological evidence.
Feature engineering and preprocessing	methodology, `src/`	Top-1,500 training-fold ingredient-presence features + log1p count; no categories/brands.
Model comparison	`reports/tables/`, final metrics	Keep the benchmark leader transparent; it may differ from the selected RF model.
Tuning and evaluation	model card, final results	Three-fold grouped training CV; internal grouped holdout, not external validation.
Visual communication	notebook, `reports/figures/`	Verify the final notebook/figures render before publishing.
Runnable code	`src/`, `requirements.txt`, `Makefile`, `Dockerfile`, CI workflow	Run locally and inspect results; Docker packaging is not a substitute for user verification.
Notebook and README	RF notebook, README	Ensure the executed RF notebook is present on the intended branch.
New GitHub repository	**User action needed**	Create/update the user's new repository and verify its default

		branch; this delivery does not push GitHub.
Slides URL	**User action needed**	Slides are delivered separately. Add the actual final URL in the required submission location; no placeholder link is published here.
Student Portal repository URL	**User action needed**	Paste the actual repository URL in Student Portal after verifying the final public tree.

Evidence snapshot from the completed run

- Selected model recorded in final metrics: **Random Forest**.
- Saved automated-test evidence: reports/test_results.txt — **23 passed**.
- Notebook execution evidence: reports/execution_summary.json — 43 notebook cells, 24 executed code cells, 0 error outputs, 15 embedded PNG figures.
- Dynamic performance evidence: reports/final_metrics.json.

Track-specific clarification

The brief makes **API use optional**; data collection is the requirement. SQL may supplement this Case Study 1 repository, but SQL plus Tableau/Power BI dashboard work is primarily associated with the alternative Case Study 2 Analytics track and is not mandatory for this selected case.

Submission integrity

The brief requires working code built by the student. Code assistance does not transfer understanding or authorship responsibility: review every relevant component, reproduce the work, make any cohort-required disclosure, and follow the cohort's AI-use policy. Assessment remains with the teaching team; this map cannot guarantee acceptance.